

Supplementary Material

Scale Fixes Local Errors First: Differential Scaling of Chess Transformer Failures

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This document holds the per-run tables and verification records that the manuscript summarises. Every table is generated from the result files in the public repository by `code/make_supplement.py`, so a table here cannot drift away from the analysis it reports. Section numbers are prefixed S and the tables are numbered S1 onward.

S1 Reproduction

The analysis runs in the following order. Each step writes result files that the next step reads, and the manuscript's numbers are generated from those files by the macro pipeline rather than typed.

1. `code/prep_data.py`, `code/make_states.py`: tokenise games and record ground-truth board state.
2. `code/run_grid.py`: train the 18 runs of the ladder.
3. `code/full_classification.py`: classify every failure at every scored position.
4. `code/regen_exponents.py`, `code/compositional.py`, `code/robustness_checks.py`: fit exponents, the closed compositional model, and the leave-one-rung-out refits.
5. `code/make_macros_ds.py`: write every number the manuscript prints, with its source file recorded beside it.
6. `code/build_ds.py`: build the manuscript and run the gates and the consistency audit.

All of it runs on one NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB of device memory under Windows 11, using Python 3.12.10, PyTorch 2.5.1+cu121, NumPy 2.4.6 and python-chess 1.11.2. The classification and analysis steps run on CPU; training used the GPU.

Table S1: Per-run training detail, read out of the checkpoints. Every run shares one recipe; the final checkpoint is also the lowest-validation-loss checkpoint in 18 of 18 runs, so checkpoint selection carries none of the result.

rung	seed	params	L	d	heads	steps	tokens (M)	val.	best
6L192	0	3,440,064	6	192	6	12,000	30.0	3.3423	3.3423
6L192	1	3,440,064	6	192	6	12,000	30.0	3.3489	3.3489
6L192	2	3,440,064	6	192	6	12,000	30.0	3.3388	3.3388
8L256	0	7,345,920	8	256	8	12,000	30.0	3.0287	3.0287
8L256	1	7,345,920	8	256	8	12,000	30.0	3.0371	3.0371
8L256	2	7,345,920	8	256	8	12,000	30.0	3.0375	3.0375
12L256	0	10,504,960	12	256	8	12,000	30.0	2.9166	2.9166
12L256	1	10,504,960	12	256	8	12,000	30.0	2.9279	2.9279
12L256	2	10,504,960	12	256	8	12,000	30.0	2.9129	2.9129
8L384	0	15,737,472	8	384	8	12,000	30.0	2.8401	2.8401
8L384	1	15,737,472	8	384	8	12,000	30.0	2.8538	2.8538
8L384	2	15,737,472	8	384	8	12,000	30.0	2.8353	2.8353
12L384	0	22,835,328	12	384	8	12,000	30.0	2.7537	2.7537
12L384	1	22,835,328	12	384	8	12,000	30.0	2.7482	2.7482
12L384	2	22,835,328	12	384	8	12,000	30.0	2.7527	2.7527
12L512	0	39,884,288	12	512	8	12,000	30.0	2.7095	2.7095
12L512	1	39,884,288	12	512	8	12,000	30.0	2.7001	2.7001
12L512	2	39,884,288	12	512	8	12,000	30.0	2.7065	2.7065

Table S2: Per-run evaluation counts. Every failure at every scored position is classified, so these are counts rather than sampled shares: 110,545 failures over 622,998 model-position evaluations, each model being evaluated at the same 34,611 unique positions. No recorded failure was a legal move under the corrected gate.

rung	seed	positions	failures	rate	empty	opp	own	cast	g.imp	g.blk	check
6L192	0	34,611	8,703	0.25145	2,164	406	1,165	41	2,006	505	2,416
6L192	1	34,611	8,830	0.25512	2,183	496	1,181	52	1,940	504	2,474
6L192	2	34,611	8,715	0.25180	2,130	430	1,230	54	1,957	528	2,386
8L256	0	34,611	6,735	0.19459	1,150	273	854	32	1,639	514	2,273
8L256	1	34,611	6,879	0.19875	1,110	271	856	27	1,771	533	2,311
8L256	2	34,611	6,825	0.19719	1,113	294	857	31	1,733	546	2,251
12L256	0	34,611	6,192	0.17890	872	221	731	20	1,639	517	2,192
12L256	1	34,611	6,174	0.17838	857	216	709	31	1,610	487	2,264
12L256	2	34,611	6,041	0.17454	777	209	709	22	1,584	515	2,225
8L384	0	34,611	5,597	0.16171	728	188	578	19	1,386	522	2,176
8L384	1	34,611	5,590	0.16151	760	257	643	15	1,336	472	2,107
8L384	2	34,611	5,433	0.15697	731	157	586	29	1,368	493	2,069
12L384	0	34,611	5,116	0.14781	590	148	500	19	1,369	476	2,014
12L384	1	34,611	5,010	0.14475	590	131	464	15	1,314	484	2,012
12L384	2	34,611	4,968	0.14354	591	144	480	17	1,268	454	2,014
12L512	0	34,611	4,586	0.13250	507	117	411	21	1,177	428	1,925
12L512	1	34,611	4,509	0.13028	497	121	389	19	1,131	445	1,907
12L512	2	34,611	4,642	0.13412	512	126	419	17	1,175	438	1,955

Table S3: Absolute failure rate per scored position at the smallest and largest rung, the fold drop between them, and the size exponent with its rung-resampled interval. The last three rows are not disjoint classes and must not be summed with the others.

class	at smallest	at largest	fold	exponent	95% CI	excl. 0
empty source	0.0624	0.0146	4.3	-0.5891	[-0.7541, -0.3847]	yes
opponent source	0.0128	0.0035	3.7	-0.5375	[-0.6129, -0.4486]	yes
own destination	0.0344	0.0117	2.9	-0.4518	[-0.5077, -0.4252]	yes
castling	0.0014	0.0005	2.6	-0.4154	[-0.5974, -0.1503]	yes
geometry, impossible	0.0569	0.0335	1.7	-0.2223	[-0.2663, -0.1810]	yes
geometry, blocked path	0.0148	0.0126	1.2	-0.0694	[-0.1183, -0.0154]	yes
leaves check	0.0701	0.0557	1.3	-0.0966	[-0.1144, -0.0851]	yes
<i>pooled local</i>	0.1665	0.0634	2.6	-0.3972	[-0.4599, -0.3184]	yes
<i>all illegal</i>	0.2528	0.1323	1.9	-0.2664	[-0.3067, -0.2160]	yes
<i>despite correct local belief</i>	0.0436	0.0395	1.1	-0.0365	[-0.0943, +0.0314]	no

Table S4: Paired differentials, bootstrapped on the same rung resamples so each interval is on the difference itself.

differential	point	95% CI	excl. 0
check - local pooled	+0.3006	[+0.2125, +0.3746]	yes
check - empty	+0.4925	[+0.2777, +0.6697]	yes
check - opponent	+0.4409	[+0.3409, +0.5183]	yes
check - geom impossible	+0.1257	[+0.1026, +0.1564]	yes
corr local - empty	+0.5526	[+0.4160, +0.6593]	yes

Table S5: The headline differential refitted with each rung deleted in turn. The sign is unchanged in every refit and the largest shift is 0.1352, so no single rung carries the result.

fit	check - empty	shift
full ladder	+0.4925	-
without 6L192	+0.3573	-0.1352
without 8L256	+0.5045	+0.0120
without 12L256	+0.5017	+0.0092
without 8L384	+0.4899	-0.0026
without 12L384	+0.4868	-0.0057
without 12L512	+0.5816	+0.0891

S2 Per-run training detail

S3 Per-run evaluation and failure counts

S4 Exponents, endpoints and fold drops

S5 Leave one rung out

S6 Locality measurements

For a position and an illegal move, k counts the squares whose contents can alone flip the legality verdict, excluding the two the move names. It is computed by brute force with no model involved, so it is a property of the rules and the position.

Table S6: Dependency set sizes over 120 games. The two geometry rows subdivide the geometry row above them; castling was sampled separately once the audit identified it as a distinct class.

class	n	mean k	median	sd	min	max
from empty	960	0.000	0	0.00	0	0
from opponent	960	0.000	0	0.00	0	0
to own	949	0.000	0	0.00	0	0
geometry	960	0.019	0	0.14	0	1
geometry impossible	932	0.000	0	0.00	0	0
geometry blocked	28	0.643	1	0.48	0	1
castling	603	0.128	0	0.45	0	7
leaves check	375	17.491	17	11.34	0	55

The statistic counts single-square sufficiency, which undercounts when no single square is sufficient. The deficit is characterised rather than unknown: among the 28 blocked-path moves, those with one blocker have $k = 1$ in 18 of 19 cases, and all 9 moves with two or more blockers have $k = 0$, since removing either blocker alone leaves the path blocked.

S7 The classifier correction, audited

Table S7: Class flow under the corrected classifier, over the 16,200 audited moves that carry both labels. No move entered or left either class the headline differential uses, so the relabelling cannot have changed it.

old label	new label	moves	
leaves check	leaves check	6,143	
geometry	geometry impossible	4,106	changed
from empty	from empty	2,327	
to own	to own	1,701	
geometry	geometry blocked	1,352	changed
from opponent	from opponent	462	
to own	castling	109	changed

Table S8: The original failure gate tested membership in the generated legal moves; the corrected gate tests legality directly. The two differ only on the king-onto-rook castling encoding. Over every scored position of every run they disagree at 0 positions, so the aggregate rate is unaffected over the whole set and not merely in a sample.

run	positions	membership gate	legality gate	disagree
6L192 s0	34,611	0.251452	0.251452	0
6L192 s1	34,611	0.255121	0.255121	0
6L192 s2	34,611	0.251799	0.251799	0
8L256 s0	34,611	0.194591	0.194591	0
8L256 s1	34,611	0.198752	0.198752	0
8L256 s2	34,611	0.197192	0.197192	0
12L256 s0	34,611	0.178903	0.178903	0
12L256 s1	34,611	0.178383	0.178383	0
12L256 s2	34,611	0.174540	0.174540	0
8L384 s0	34,611	0.161712	0.161712	0
8L384 s1	34,611	0.161509	0.161509	0
8L384 s2	34,611	0.156973	0.156973	0
12L384 s0	34,611	0.147814	0.147814	0
12L384 s1	34,611	0.144752	0.144752	0
12L384 s2	34,611	0.143538	0.143538	0
12L512 s0	34,611	0.132501	0.132501	0
12L512 s1	34,611	0.130277	0.130277	0
12L512 s2	34,611	0.134119	0.134119	0

Table S9: Why the headline differential moved, varying one thing at a time. The dominant term is the estimator, not the hardware: the sampled estimator was biased by corpus order. The classifier term is exactly zero, which independently confirms the flow table.

configuration	classifier	arithmetic	estimator	check – empty	shift
A published	old	float16 GPU	sampled	+0.4613	–
B precision	old	float32 CPU	sampled	+0.4561	-0.0052
C estimator	old	float32 CPU	full set	+0.4925	+0.0364
D classifier	new	float32 CPU	full set	+0.4925	+0.0000

The decomposition of the near-flat joint rate is exact rather than estimated, and that commits it to a checkable property. Since the identity holds position by position, the logs add and an ordinary-least-squares slope is a linear functional of the response, the three exponents must sum. They do, to 2.4e-16: +0.1479 for the marginal and -0.1844 for the conditional give -0.0365 for the joint.

S8 What the verification suite checks

The build fails rather than warns. Two kinds of check run on every build: gates on the manuscript, and invariants between quantities that must agree. Each was added after a defect of its kind reached a draft.

- **Provenance.** Every number in the manuscript resolves to a macro with a recorded source file, no decimal may be typed in prose, and no macro sourced from the superseded classification pass may appear at all.
- **De-escaped control sequences.** Raw tabs and carriage returns in the source, and orphaned fragments such as `exttt`, which typeset as visible prose while the document still compiles.
- **Arithmetic.** The partition sums to the illegal-move rate; fold drops agree with their end-points; relabelling counts sum to their components; the exponents of an exact decomposition sum to the exponent they decompose.
- **Significance.** No class may be described as having an interval that includes zero when its interval excludes it.
- **Scope.** Every cross-reference resolves; every quoted count is no larger than its total; the anonymous build contains no author identity; the source archive compiles from its own extracted contents.

S9 Artifact inventory

Table S10: Result files cited by the manuscript or this supplement. All are in the public repository.

file under <code>results/</code>	contents
<code>full_classification.json</code>	every failure classified, per run
<code>regen_exponents.json</code>	exponents, shares, depth bands, differentials
<code>attn_*_regen.json</code>	the audited sample, each move under both labels
<code>aggregate_gate_check.json</code>	both legality gates on every position
<code>drift_decomposition.json</code>	why the headline differential moved
<code>locality_index.json</code>	dependency set sizes by class
<code>blocker_check.json</code>	the multi-blocker characterisation of k
<code>training_table.json</code>	per-run architecture and training detail
<code>compositional.json</code>	the closed simplex fit and the closure argument
<code>robustness_checks.json</code>	leave-one-rung-out refits
<code>joint_vs_conditional.json</code>	the exact decomposition and its additivity
<code>environment.json</code>	hardware and library versions
<code>prereg_timing.json</code>	registration dates read from the git history